

Leonard E. van Dyck

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EDUCATION

PhD in Psychology (Computational Cognitive Neuroscience)

Justus Liebig University Giessen

Advisors: Prof. Katharina Dobs & Prof. Martin Hebart

Giessen, Germany

09/2023 - present

MSc in Psychology (Cognitive Neuroscience)

University of Salzburg

Advisors: Prof. Mario Braun & Prof. Martin Kronbichler

Salzburg, Austria

10/2020 - 02/2023

BSc in Psychology

University of Salzburg

Advisor: Dr. Walter Gruber

Salzburg, Austria

10/2017 - 07/2020

EXPERIENCE

Guest Researcher

Max Planck Institute for Human Cognitive and Brain Sciences

Group Leader: Prof. Martin Hebart

Leipzig, Germany

10/2023 - 09/2025

Laboratory Manager

Centre for Cognitive Neuroscience, University of Salzburg

Method Unit EEG

Salzburg, Austria

10/2020 - 02/2023

Teaching Assistant

Department of Psychology, University of Salzburg

Bachelor's Seminar "Artificial Intelligence"

Salzburg, Austria

03/2021 - 07/2022

Research Assistant

Laboratory for Sleep, Cognition, and Consciousness Research,

University of Salzburg

Group Leaders: Prof. Manuel Schabus & Prof. Kerstin Hödlmoser

Salzburg, Austria

03/2020 - 07/2020

Research Internship

Department of Psychology, University of Salzburg

Clinical and Family Therapy, Prof. Katherine Hertlein

(University of Nevada, Las Vegas)

Salzburg, Austria

04/2019 - 07/2019

CONTRIBUTIONS

[Google Scholar profile \(up to date\)](#), *equal contribution

Preprints

2. **van Dyck, L. E.** & Dobs., K. (2026). Face and body representations converge along the visual hierarchy in models and cortex. *bioRxiv*. <https://doi.org/10.64898/2026.02.16.706115>
1. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2025). Multidimensional feature tuning in category-selective areas of human visual cortex. *bioRxiv*. <https://doi.org/10.1101/2025.06.17.659578>

Publications

6. **van Dyck, L. E.** & Dobs, K. (2025). Category selectivity as a window into behavioral relevance. *Cognitive Neuroscience*, 1-3. <https://doi.org/10.1080/17588928.2025.2590656>
5. **van Dyck, L. E.**, Bremmer, F., & Dobs, K. (2024). Artificial intelligence meets body sense: task-driven neural networks reveal computational principles of the proprioceptive pathway. *Signal Transduction and Targeted Therapy*, 9(1), 171. <https://doi.org/10.1038/s41392-024-01870-9>
4. **van Dyck, L. E.** & Gruber, W. R. (2022). Modeling biological face recognition with deep convolutional neural networks. *Journal of Cognitive Neuroscience*, 35(10), 1521–1537. https://doi.org/10.1162/jocn_a_02040
3. **van Dyck, L. E.**, Denzler, S. J., & Gruber, W. R. (2022). Guiding visual attention in deep convolutional neural networks based on human eye movements. *Frontiers in Neuroscience*, 16. <https://doi.org/10.3389/fnins.2022.975639>
2. **van Dyck, L. E.**, Kwitt, R., Denzler, S. J., & Gruber, W. R. (2021). Comparing object recognition in humans and deep convolutional neural networks – An eye tracking study. *Frontiers in Neuroscience*, 15. <https://doi.org/10.3389/fnins.2021.750639>
1. Hertlein, K. M. & **van Dyck, L. E.** (2020). Predicting engagement in electronic surveillance in romantic relationships. *Cyberpsychology, Behavior, and Social Networking*, 23(9), 604-610. <https://doi.org/10.1089/cyber.2019.0424>

Conference Presentations

9. **van Dyck, L. E.** & Dobs., K. (2025). Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex. **Poster** presented at *Cognitive Computational Neuroscience (CCN)*, August 12-15, Amsterdam, NL. [Short paper](#)
8. **van Dyck, L. E.** & Dobs., K. (2025). Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex. **Poster** presented at *Workshop on CONCEPTS, ACTIONS, and OBJECTS (CAOs)*, May 7-9, Rovereto, Italy.
7. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. **Poster** presented at *Neuro-AI-Talks (NEAT)*, September 02-03, Osnabrück, Germany.
6. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. **Poster** presented at *European Conference on Visual Perception (ECPV)*, August 25-29, Aberdeen, Scotland.
5. **van Dyck, L. E.**, Hebart, M. N., & Dobs., K. (2024). Core neural dimensions of functionally selective areas in the human visual cortex. **Poster** presented at *Cognitive Computational Neuroscience (CCN)*, August 6-9, Boston, MA, USA. [Short paper](#)

4. **van Dyck, L. E.**, Hebart, M. N., & Dobs., K. (2024). Neural representational dimensions capture the nested functional organization of the human visual cortex. **Talk** presented at *SFB Workshop Categorization in Perception and Action: Minds, Models, Mechanisms*, June 30 - July 2, Marburg, Germany.
3. **van Dyck, L. E.**, Dobs., K.* , & Hebart, M. N.* (2024). Data-driven voxel decomposition reveals representational dimensions in functionally selective areas. **Poster** presented at *Workshop on CONCEPTS, ACTIONS, and OBJECTS (CAOs)*, May 9-11, Rovereto, Italy.
2. **van Dyck, L. E.**, Denzler, S. J., & Gruber, W. R. (2022). Analyzing and increasing the similarity of humans and deep convolutional neural networks in object recognition. **Poster** presented at *European Conference on Visual Perception (ECVP)*, August 28 - September 1, Nijmegen, The Netherlands.
1. **van Dyck, L. E.**, Denzler, S. J., Schöllkopf, C. P., & Gruber, W. R. (2022). Spatial similarities between human eye movements and deep convolutional neural network saliency maps across time. **Poster** presented at *Salzburg Mind Brain Annual Meeting (SAMBA)*, July 14-15, Salzburg, Austria.

Theses

- van Dyck, L. E.** (2023). Unraveling top-down and bottom-up processes in theory of mind with layer fMRI. [Master's Thesis]. <https://eplus.uni-salzburg.at/obvusbhs/content/titleinfo/8960559>
- van Dyck, L. E.** & Gruber, W. R. (2020). Seeing Eye-to-Eye? A comparison of object recognition performance in humans and deep convolutional neural networks under image manipulation. arXiv [Bachelor's Thesis]. <https://doi.org/10.48550/arxiv.2007.06294>

Invited Talks

1. *Two Approaches to Increasing the Human-Likeness of Visual Attention in DCNNs*
Vision and Computational Cognition Group (Group Leader: Prof. Martin Hebart)
Max Planck Institute for Human Cognitive and Brain Sciences, September 29, 2022

Science Communication

- van Dyck, L. E.** & Dobs, K. (2023). Modellierung der biologischen Gesichtswahrnehmung mit Künstlicher Intelligenz. *DER AUGENSPIEGEL* (German Journal for Ophthalmologists), December 2023, 50-53. <https://www.augenspiegel.com/modellierung-der-biologischen-gesichtswahrnehmung-mit-kuenstlicher-intelligenz/>

GRANTS AND AWARDS

Doctoral Scholarship Studienstiftung des deutschen Volkes e.V. German Academic Scholarship Foundation	10/2023 - present
Merit Scholarship University of Salzburg	2023
ECVP22 Travel Grant Donders Institute for Brain, Cognition, and Behaviour	2022

ACTIVITIES AND SERVICE

Ad-hoc Reviewer

The Journal of Neuroscience, PLOS Computational Biology, Frontiers in Neuroscience, British Journal of Psychology, Cognitive Computational Neuroscience Conference

SKILLS

Programming

- Languages: Python, MATLAB, R, Bash
- Machine Learning: Scikit-learn, PyTorch
- Data Analysis: NumPy, Pandas, Matplotlib, Jupyter
- Version Control: Git

Laboratory

- Experiment Design: PsychoPy, PsychToolbox
- Data Acquisition: EEG, fMRI, Eye Tracking, Psychophysics, Behavior
- Data Analysis: EEG/MEG (FieldTrip), fMRI (SPM, FSL, FreeSurfer, Nilearn, PyCortex), Eye Tracking, Psychophysics, Behavior

Languages

- German (Native)
- English (Proficient - C1)
- French (Intermediate - B1)